Lecture-17
Differential Analysis of fluid motion I: Conservation of mass

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Learning Objectives

By the end of this lecture, you should be able to

1. Use the **Gauss divergence theorem** to convert the **integral conservation of mass equation to its differential form**
2. Derive the conservation of mass equation by applying **mass balance on a differential fluid element**
3. Reduce the general equation to **specialized forms for steady and incompressible flow fields**
4. Do **some engineering applications** of the conservation of mass equation

COM from the integral form

- Recall that the **conservation of mass equation for a control volume** could be expressed as follows

$$\frac{\partial}{\partial t}\int_{CV}\rho\, d\forall + \int_{CS}\rho\vec{V}\cdot d\vec{A} = 0$$

- Also recall that from the Gauss divergence theorem

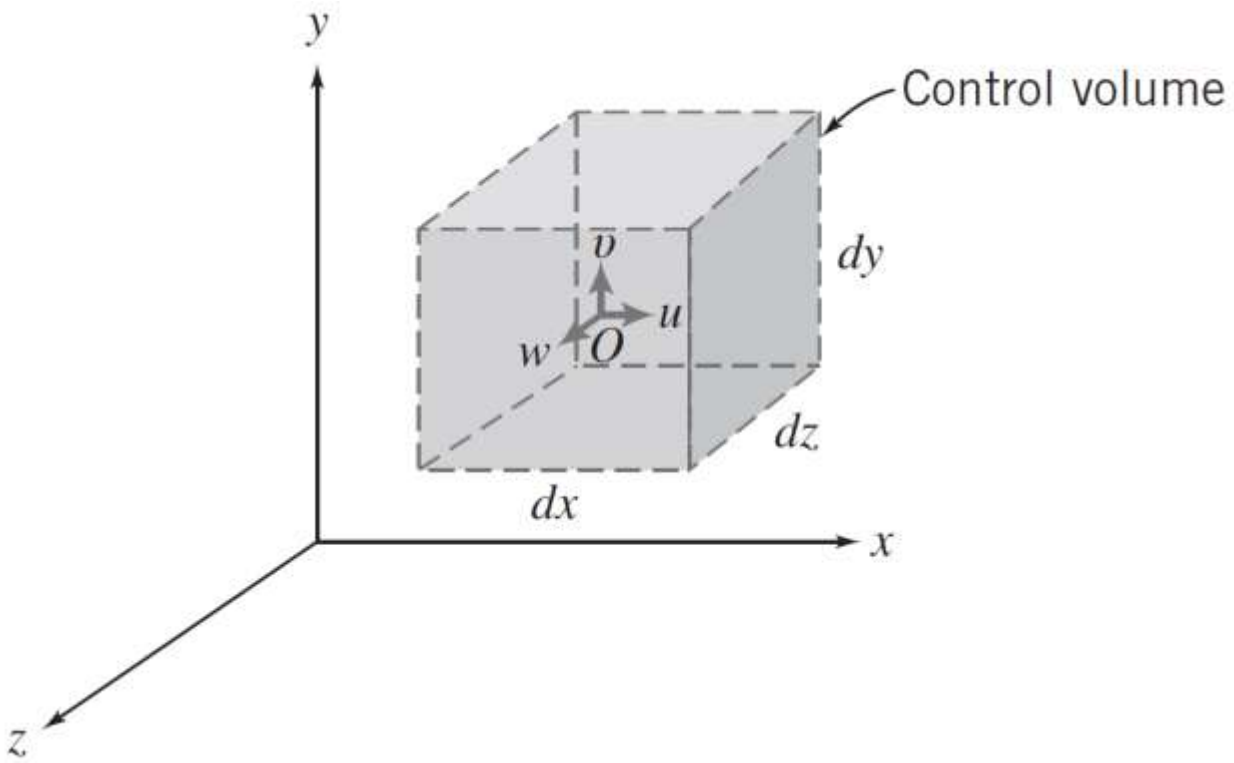
$$\iint_{S(V)} \mathbf{F}\cdot\hat{\mathbf{n}}\,\mathrm{d}S = \iiint_{V} \operatorname{div}\mathbf{F}\,\mathrm{d}V$$

- Since the choice of a control volume is arbitrary, we get

$$\nabla\cdot\rho\vec{V} + \frac{\partial\rho}{\partial t} = 0$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.
https://en.wikipedia.org/wiki/Divergence_theorem

COM from a bottom-up approach



$$\frac{\partial}{\partial t}\int_{CV}\rho\, d\forall + \int_{CS}\rho\vec{V}\cdot d\vec{A} = 0$$

$$\int_{CS}\rho\vec{V}\cdot d\vec{A} = \left[\frac{\partial \rho u}{\partial x} + \frac{\partial \rho v}{\partial y} + \frac{\partial \rho w}{\partial z}\right] dx\, dy\, dz$$

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Revisiting the learning objectives

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